

46 YEARS OF EXCELLENCE

NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR.IIT_*CO-SC(MODEL-A& APEX) **Date:27-12-24**

Time: 3HRS **GTM-4** **Max. Marks: 300**

PHYSICS **MAX.MARKS: 100**

SECTION – I

(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. The density of a material in the shape of a cylinder is determined by measuring its length, radius & mass. If the relative error in measuring mass, length, radius are 1%, 2%, 1.5% respectively, then the maximum error in determining the density is

   A) 6% B) 4.5% C) 3% D) 2.5%

2. In a series LCR circuit, the inductive reactance is 10Ω and capacitive reactance is 2Ω. The resistance R in the circuit is 6Ω. The power factor of the circuit is_____

   A) $\frac{5}{3}$ B) $\frac{3}{5}$ C) $\frac{2}{5}$ D) $\frac{1}{3}$

3. An artificial satellite moving in a circular orbit around the earth has a total (K.E+P.E) energy E_0. Its potential energy is

   A) $-E_0$ B) $1.5\ E_0$ C) $2\ E_0$ D) E_0

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4. A liquid of density 750 kgm^{-3} flows smoothly through a horizontal pipe whose cross sectional area changes from $1\times10^{-2}m^2$ to $0.5\times10^{-2}m^2$. The pressure difference between wide & narrow sections of the pipe is 4500 pascal. Velocity of liquid at wide section is _____

A) 1 m/s B) 2 m/s C) 4 m/s D) 0.5 m/s

5. Let F_{pp}, F_{pn} & F_{nn} denote the magnitudes of the net force by a proton on a proton, by a proton on a neutron and by a neutron on a neutron respectively, neglect gravitational force. Then

A) $F_{pp} > F_{pn} = F_{nn}$ B) $F_{pp} = F_{pn} = F_{nn}$ C) $F_{pp} > F_{pn} > F_{nn}$ D) $F_{pp} < F_{pn} = F_{nn}$

6. In Rutherford's α ray scattering experiment distance of closest approach for the α particle is $9\times10^{-14}m$. If the target nucleus has atomic number 80, then maximum K.E of α particle in electron volt is_____ ($\frac{1}{4\pi\in_0}=9\times10^9$ SI unit)

A) 256×10^4 B) 16×10^4 C) 128×10^4 D) 4096×10^4

7. In constant volume gas thermometer, pressure of gas at $0^0C, 100^0C$ and θ^0C is found to be 1 atm, 2atm, 1.6 atm respectively. The temperature θ in ^0C is ____

A) 20^0C B) 30^0C C) 50^0C D) 60^0C

8. Two particles of mases 2kg and 3kg are moving in perpendicular directions with velocities 4m/s and 2m/s respectively. The magnitude of velocity of centre of mass of two particles is _____

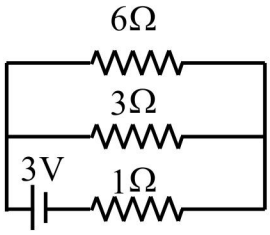
A) 1 m/s B) 2.8 m/s C) 0.4 m/s D) 2 m/s

9. The one division of main scale of vernier callipers reads 1mm and 10 divisions of vernier scale is equal to the 9 divisions on main scale. When the two jaws of the instrument touch each other the zero of the vernier lies to right of zero of the main scale and its fourth division coincides with a main scale division. When a spherical bob is tightly placed between the two jaws, the zero of the vernier scale lies in between 4.1 cm and 4.2 cm and 6th vernier division coincides with a main scale division. The diameter of the bob is _____

A) 4.12 cm B) 4.20 cm C) 4.16 cm D) 4.18 cm

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10. In the given circuit current flowing through 3Ω resistance is____



A) 1 A B) $\frac{1}{2}A$ C) $\frac{1}{3}A$ D) $\frac{2}{3}A$

11. Average force exerted on a reflecting surface at normal incidence is $2\times10^{-4}N$. If $300\,W/cm^2$ is the light energy flux during span of 2hours, then area of the surface is

A) $0.02m^2$ B) $0.01m^2$ C) $0.03m^2$ D) $0.04m^2$

12. 2 moles of mono atomic gas is mixed with 4 moles of diatomic gas (at room temperature) the value of $\frac{C_p}{C_v}$ for the mixture is

A) $\frac{16}{13}$ B) $\frac{13}{19}$ C) 2 D) $\frac{19}{13}$

13. If kinetic energy of a body becomes four times. Then the percentage change in its momentum is_____

A) 41.4 % B) 200 % C) 100 % D) 400 %

14. A bomb of mass m at rest explodes into 2 pieces of masses in the ratio 1:3. If speed of lighter piece is v after explosion, then the kinetic energy just after explosion of the bomb is_____

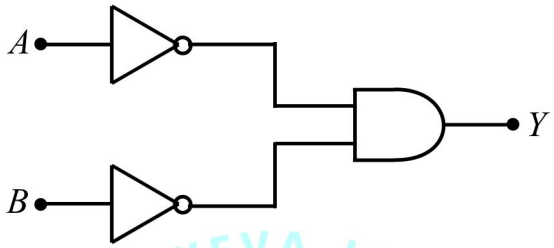
A) $\frac{2}{3}mv^2$ B) $\frac{mv^2}{2}$ C) $\frac{mv^2}{3}$ D) $\frac{mv^2}{6}$

15. A rigid rod rotates about its one end in horizontal circle. If speed of particle on the rod at a distance of 2cm from its axis is 6m/s then the speed of another particle on the rod at a distance of 3cm from the same axis is_____

A) 2 m/s B) 6 m/s C) 9 m/s D) 3 m/s

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16. The truth table for the circuit given in the figure is



A)
A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

B)
A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

C)
A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

D)
A	B	Y
0	0	1
0	1	1
1	0	1
1	1	1

17. A force (F) acting on a body at rest varies with time t as $F=(4t+1)$ newton. The momentum of the body at time $t=3$ second is____ $\left(\text{in } kg\,ms^{-1}\right)$

A) 39 B) 14 C) 19 D) 21

18. Two conducting spheres A and B having charges $7\mu C$ and $-1\mu C$ are kept in contact, then the charge on sphere A becomes ____ (ratio of radii of spheres A to B is 1 : 2)

A) 1 μC B) 2 μC C) 3 μC D) 4 μC

19. Angle of incidence of light in medium 'A' is 45^0, when it enters into medium B then ray deviates towards normal by 15^0. The ratio of velocity of light in medium A to that of medium B is____

A) $1:\sqrt{2}$ B) $1:2$ C) $\sqrt{3}:\sqrt{2}$ D) $\sqrt{2}:1$

20. An electron at rest is accelerating through a potential difference of 300 volt. If the de Broglie wavelength of electron in $\overset{0}{\text{A}}$ is $\frac{1}{\sqrt{m}}$, then m is_____

A) 1 B) 2 C) 4 D) 3

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SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

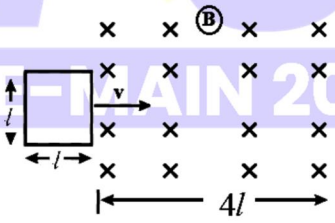
This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme: Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

21. A closed and open organ pipes in first overtone produces 4 beats per second. If velocity of sound in air 330m/s, length of open and closed organ pipes are 30cm and $\frac{n}{2208}cm$ respectively. The value of n is____ (frequency in closed pipe is greater than frequency in open pipe)

22. A proton goes undeflected in a crossed electric and magnetic filed (the fields are perpendicular to each other) at a speed of $4\times10^5 m/s$. The velocity is perpendicular to both the fields. When the electric field is switched off. The proton moves along a circle of radius 2cm. If the magnitude of electric field strength is $P\times10^4 V/m$. Then value of P is _____ ($m_P = 1.6\times10^{-27} kg$)

23. A liquid drop of radius 2 cm is sprayed into 64 droplets of equal size. If the surface tension of liquid is 0.4 N/m. The energy change is $48\pi k\times10^{-5} J$, where k is _____

24. A square loop of edge l and resistance R enters into a uniform magnetic field B at time $t=0$. The loop crosses the magnetic field with constant velocity v. If the heat produced in the loop is $\frac{kB^2 v l^3}{R}$, the value of k is _____



25. An electric field $\vec{E} = -\left(2x\hat{i} + 3y^2\hat{j}\right) V/m$ exists in space. If the potential at the origin is taken to be zero, the potential at $(1m, 2m)$ is kV, the value of k is _____

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CHEMISTRY **MAX.MARKS: 100**

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26. $CaCO_3(s) \rightleftharpoons Ca^{2+}(aq) + CO_3^{2-}(aq)$

The molar solubility product constant of $CaCO_3$ is 10^{-4} at a certain temperature.

Choose the correct statement from the following.

A) The molarity of its saturated solution is $10^{-4}\,M$

B) The solubility increases by adding HCl

C) The value of K_{SP} decreases by adding common ion

D) Precipitation of $CaCO_3$ takes place if 200 mL of 0.001 M $CaCl_2$ solution is mixed with 800 mL of 0.001 M Na_2CO_3

27. Which of the following compound does not evolve a gas with dilute H_2SO_4

A) K_2CO_3 B) K_2S C) KNO_2 D) KNO_3

28. $CH_3—C\equiv C—CH_3 \xrightarrow{Na/\text{liquid } NH_3} (P)$

$(P) \xrightarrow{MnO_4^-/OH^-/Cold} R + S$

$(P) \xrightarrow{Br_2/CCl_4} (Q)$

Which of the following statement is correct about the products of above reaction?

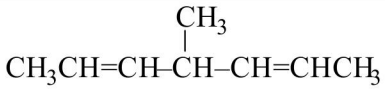
A) 'Q' contains no asymmetric carbon atom

B) 'P' and 'Q' are functional isomers

C) 50% of R and 50% of Q makes a racemic mixture

D) 'P' has one stereoisomer

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29. 

How many stereoisomers is possible for the above compound?

A) 2 B) 3 C) 4 D) 6

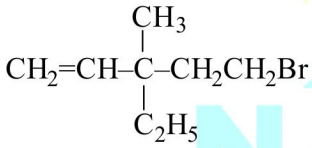
30. 2 moles of a non-volatile solute was dissolved in 8 moles of a solvent. If P^o is the vapour pressure of the pure solvent and P_S is the vapour pressure of the solution, then the correct relationship is

A) $P^o - P_S = \frac{8}{10}$ B) $\frac{P^o}{P_S} = \frac{5}{4}$ C) $P^o \div P_S = \frac{4}{5}$ D) $P^o \times P_S = \frac{10}{8}$

31. Which of the following substance can decrease the ionization of CH_3COOH in water?

$CH_3COOH(l) \rightleftharpoons CH_3COO^-(\text{aq}) + \text{H}^+(aq)$

A) $NaOH$ B) NH_4Cl C) K_2CO_3 D) $NaCl$

32. 

Which of the following substance converts the above optically active compound to optically inactive compound?

A) HBr B) Br_2 / CCl_4 C) HBr / R_2O_2 D) Br_2 / H_2O

33. Which of the following orbital has the maximum number of radial nodes?

A) 4s B) 4p C) 4d D) 4f

34. The composition of an organic compound is given below

Elements	Mass%
C	67.3
H	4.67
N	13.08
O	14.95

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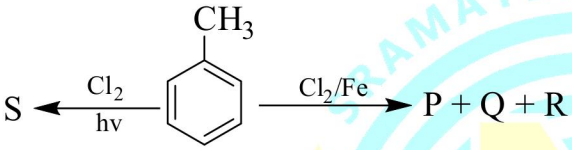
What is the approximate empirical mass of the compound in g mol^{-1} unit?

A) 170 B) 107 C) 177 C) 182

35. Rates of the following nucleophiles towards the reaction with *MeBr* in ethanol follow the order

A) $NH_3 < PhO^- < EtO^- < PhS^-$ B) $NH_3 < PhO^- < PhS^- < EtO^-$

C) $NH_3 < PhS^- < EtO^- < PhO^-$ D) $NH_3 < EtO^- < PhS^- < PhO^-$

36. 

Choose the correct statement from the following?

A) Formation of P and S follow the same mechanism.

B) 'P' and 'Q' are functional isomers.

C) 'P' and 'S' will form the same product when treated with aq.KOH.

D) In P, Q and R, one of the products, on hot alkaline permanganate oxidation, form m-chloro benzoic acid as the product.

37. What will be the oxidation potential (nearly) of the following hydrogen electrode at 298 K ? (Given: $\frac{2.303RT}{F} = 0.0591$)

$Pt|H_2(1atm)||HCl(pH=4)$

A) 0.014 V B) – 0.2364 V C) – 0.014 V D) 0.2364 V

38. What is the sum of the number of atoms which are present in a molecule of pyrogallol ?

A) 16 B) 15 C) 10 D) 14

39. The entropy of surrounding is greater than that of the system in a/an

A) irreversible adiabatic compression B) irreversible isothermal compression

C) irreversible adiabatic expansion D) irreversible isothermal expansion

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40. At a certain temperature, the standard heats of formation of $SO_{2(g)}$ and $SO_{3(g)}$ are –x and –y kcal mol^{-1}. The enthalpy change of the reaction $2SO_{2(g)} + O_{2(g)} \rightarrow 2SO_{3(g)}$ can be expressed as

A) $\left(\frac{y-x}{2}\right) K\,Cal\,mol^{-1}$ B) $(2y-2x)\,K\,Cal\,mol^{-1}$

C) $\left(\frac{x-y}{2}\right) K\,Cal$ D) $(2x-2y)\,K\,Cal$

41. Which molecule has maximum bond angle ?

A) H_2S B) H_2Se C) H_2Te D) OF_2

42. Which of the following alkyl halide forms the maximum number of hydrocarbons as products upon reaction with alc. KOH?

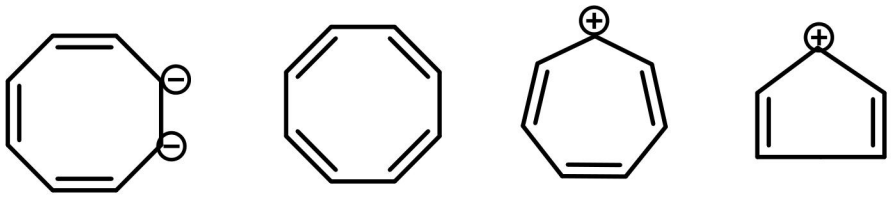
A) $CH_3CH_2CH_2CH_2Cl$ B) $CH_3CH_2\underset{\displaystyle Cl}{\underset{|}{C}}HCH_3$

C) $ClCH_2CH_2CH_2CH_2Cl$ D) $CH_3-\underset{\displaystyle CH_3}{\underset{|}{\overset{\displaystyle CH_3}{\overset{|}{C}}}}-Cl$

43. 0.7 gram of an organic compound upon Kjeldahl's estimation requires 30 ml of 0.5 M H_2SO_4 for its complete neutralization. The percentage of nitrogen in the organic compound is ____

A) 30 B) 45 C) 15 D) 60

44. Among the following, which option is correct related to following species?



(P) (Q) (R) (S)

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A) 'P' and 'Q' are non-aromatic

B) 'Q' and 'S' are anti-aromatic

C) 'Q' is anti-aromatic and 'R' is aromatic

D) 'P' is an aromatic and 'Q' is non-aromatic

45. Which of the following order is incorrect?

A) $F_2 < Cl_2 < Br_2 < I_2$ [Bond length]

B) $F_2 > Cl_2 > Br_2 > I_2$ [Oxidizing power]

C) $F_2 > Cl_2 > Br_2 > I_2$ [Bond dissociation energy]

D) $F_2 < Cl_2 < Br_2 < I_2$ [Boiling point]

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46. According to crystal field theory, the total number of unpaired electrons present in the following complex ions $[NiCl_4]^{2-}, [Ni(CN)_4]^{2-}$ and $[Ni(H_2O)_6]^{2+}$ are

47. How many of the following statements are correct ?

(i) $Cl > F$ (Negative electron gain enthalpy)

(ii) $Si > C$ (Negative electron gain enthalpy)

(iii) $Al > In$ (Electropositive character)

(iv) $Pb > Sn$ (Ionisation energy)

(v) $O > S$ (maximum oxidation state possible)

(vi) $N > P$ (maximum covalency)

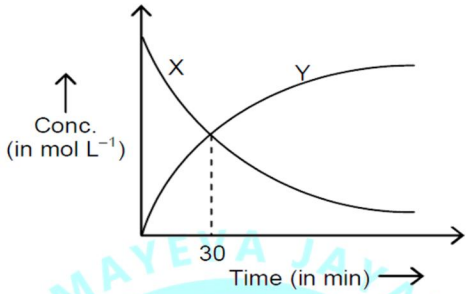
48. How many of the following species is/are paramagnetic in nature?

$O_2, O_2^+, O_2^{2-}, C_2, B_2, N_2^+, O_2^-, O_2^{2+}$

49. $31X(g) \longrightarrow Y(g)$

The following plot is given for above first order reaction. What is $t_{1/2}$ of the reaction in minutes

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50. Solid X establishes the following equilibrium at 298 K in a closed vessel.

$X(s) \rightleftharpoons Y(g) + 2Z(g)$

The equilibrium pressure is 6 atm at same temperature. The equilibrium constant (K_P) of the reaction at 298 K is_____

MATHS **MAX.MARKS: 100**

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(SINGLE CORRECT ANSWER TYPE)

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51. Let the curve $y = y(x)$ be the solution of the differential equation, $\frac{dy}{dx} = 2(x+1)$. If the area bounded by the curve $y = y(x)$ and x-axis is $\frac{8\sqrt{2}}{3}$, then the value of $y(2)$ is equal to

A) -6 B) 7 C) 9 D) 8

52. If the circles $x^2 + y^2 + 6x + 8y + 16 = 0$ and $x^2 + y^2 + 2(3-\sqrt{3})x + 2(4-\sqrt{6})y = k + 6\sqrt{3} + 8\sqrt{6}$, $k > 0$ touch internally at the point $P(-\sqrt{3}-\lambda, -\mu-\sqrt{6})$, where $\lambda, \mu \in N$, then $2\lambda + \mu =$

A) 10 B) 11 C) 9 D) 7

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53. The sum of all the real roots of the equation $\left(e^{2x}-4\right)\left(6e^{2x}-5e^{x}+1\right)=0$ is

A) $\log_e 3$ B) $\log_e\left(\frac{1}{3}\right)$ C) $\log_e 6$ D) $\log_e\left(\frac{1}{6}\right)$

54. If the domain of the function $f(x)=\frac{1}{\sqrt{4x^2+11x+6}}+\cos^{-1}(4x+3)+\sin^{-1}\left(\frac{10x+6}{3}\right)$ is $(\alpha,\beta]$, then $36|\alpha+\beta|$ is equal to

A) 45 B) 72 C) 54 D) 63

55. The equations of two sides PQ and PR of a triangle PQR are $3x-2y=5, 4x+y=14$ respectively. The point $\left(2,-\frac{4}{3}\right)$ divides the third side QR internally in the ratio 1 : 2. Equation of the third side QR is_____

A) $x+6y+6=0$ B) $x+3y+2=0$ C) $x-6y-10=0$ D) $x-3y-6=0$

56. Let $\overline{p}=4\hat{i}-\hat{j}+\hat{k}, \overline{q}=11\hat{i}-\hat{j}+\hat{k}$ and $\overline{r}$ be a vector such that $(\overline{p}+\overline{q})\times\overline{r}=\overline{r}\times(-2\overline{p}+3\overline{q})$. If $(2\overline{p}+3\overline{q}).\overline{r}=1670$, then $|\overline{r}|^2$ is equal to

A) 1600 B) 1618 C) 1627 D) 1609

57. The lines $\frac{x-1}{2}=\frac{y-z}{1}=\frac{z-1}{2}$ and $\frac{x-5}{2}=\frac{y-p}{-1}=\frac{z-5}{3}$ are intersecting at $Q(a,b,c)$. Then $a+b+c+p$ is

A) 14 B) 18 C) 24 D) 20

58. Let $A=\begin{bmatrix}2&1&2\\6&2&11\\3&3&2\end{bmatrix}$ and $P=\begin{bmatrix}1&2&0\\5&0&2\\7&1&5\end{bmatrix}$. Number of prime factors of $\left|P^{-1}AP-2I\right|$ is equal to

A) 26 B) 69 C) 3 D) 2

59. Let $f(x)=(x+3)^2(x-2)^3, x\in[-4,4]$. If M and m are the maximum and minimum values of $f(x)$ respectively in $[-4,4]$, then the value of $M+m$ is

A) – 608 B) – 176 C) 176 D) 608

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60. The sum of the series $\frac{{}^{100}C_{25}}{100}+\frac{{}^{99}C_{25}}{99}+\frac{{}^{98}C_{25}}{98}....+\frac{{}^{50}C_{25}}{50}$ is_____

A) $\frac{1}{25}\left[{}^{100}C_{25}+{}^{50}C_{25}\right]$ B) $\frac{1}{25}\left[{}^{100}C_{25}-{}^{51}C_{25}\right]$

C) $\frac{1}{25}\left[{}^{100}C_{25}-{}^{49}C_{25}\right]$ D) $\frac{1}{25}\left[{}^{100}C_{25}-{}^{50}C_{26}\right]$

61. If $f(x)=\sqrt{\cos^2 x}+\sqrt{\sin^2 x}$ *then* $\int_{-3/2}^{3/2}\left[f(x)\right]dx=$ (where [.] is G.I.F.)

A) 3 B) 4 C) 5 D) 6

62. An ellipse and a hyperbola having centre at origin and foci on $(\pm 10, 0)$. If the distance between their directrices (both in positive direction of x axis) is 10 units, then positive difference of square of length of their major axis and transverse axis is_____

A) 400 B) 300 C) 200 D) 100

63. Let $z_1=3+4i$ and $z_2=2+i$ and z satisfies the equation $2\left(z+\bar{z}\right)+3\left(z-\bar{z}\right)i=0$. If $|z-z_1|+|z-z_2|$ is minimum, then possible value of z can be

A) $\frac{15}{7}+\frac{10}{7}i$ B) $\frac{15}{7}+\frac{8}{7}i$ C) $\frac{15}{7}+\frac{9}{7}i$ D) $\frac{10}{7}+\frac{15}{7}i$

64. If $\left(x+\sqrt{1+x^2}\right)\left(y+\sqrt{1+y^2}\right)=1$, then $\frac{dy}{dx}$ may be equal to____

A) 0 B) –1 C) 1 D) 2

65. A survey shows that in a city, 60% families own a car, 80% families have a scooter, and 40% have a bicycle. Also 30% own both a car and scooter, 35% car and bicycle and 25% scooter and bicycle and some families owns all the three. Now the families who have neither of the three can be

A) 7% B) 11% C) 12% D) None of these

66. $\left(1^2-2^2+3^2\right)+\left(4^2-5^2+6^2\right)+\left(7^2-8^2+9^2\right)....$ upto 30 terms is___

A) 82395 B) 80396 C) 82389 D) 82399

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67. If $\frac{\cos x}{\cos\theta}+\frac{\sin x}{\sin\theta}=\frac{\cos y}{\cos\theta}+\frac{\sin y}{\sin\theta}=1$ (where $x-y\neq n\pi, n\in Z$),

then $\frac{\cos x\cos y}{\cos^2\theta}+\frac{\sin x\sin y}{\sin^2\theta}=$

A) $\frac{1}{2}$ B) 1 C) -1 D) $\frac{3}{2}$

68. If the tangents drawn from $(a,0)$ to $y^2=4a(x+a^2)$ are at right angles, then 'a' can be

A) 1 B) –2 C) 3 D) –3

69. Number of ways in which all the four letters L_1,L_2,L_3,L_5 can be placed in five envelopes E_1,E_2,E_3,E_4,E_5 such that each envelope can contain a maximum of one letter and letter L_i should not be placed in envelope E_i $(i=1,2,3,5)$ is

A) 65 B) 51 C) 44 D) 53

70. If $f(n)=\begin{vmatrix} n^2 & (n+1)^2 & (n+2)^2 \\ (n+1)^2 & (n+2)^2 & (n+3)^2 \\ (n+2)^2 & (n+3)^2 & (n+4)^2 \end{vmatrix}$ (where $n\in N$), then

A) $f(5)=-125$ B) $f(6)=-216$ C) $f(2)=-8$ D) $f(1)=-1$

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme: Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

71. Given $y(0)=2000$ and $\frac{dy}{dx}=32000-20y^2$, then find the value of $\lim_{x\to\infty} y(x)$.

72. In a series of seven T–20 matches between India and England, their respective winning probabilities is $\frac{2}{5}$ and $\frac{3}{5}$. If probability of India winning 'n' matches is highest, then 'n' is_____

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73. If $[a,b]$ is the range of the function $f(x)=\sin^{-1}x+\cos^{-1}x+\tan^{-1}x$, then the number of roots of the equation $1-|x|=\tan^{-1}x$ is $([b]+k)$. Value of k is ____ (where [.] is G.I.F.)

74. Let $x_1,x_2,.....,x_{100}$ *are* 100 observation such that $\sum_{i=1}^{100}x_i=0$, $\sum_{1\le i<j\le 100}|x_i.x_j|=80000$ and mean deviation from their mean is 5. Value of standard deviation is_____

75. If $f(x)=\begin{cases} x\left(\dfrac{ae^{\frac{1}{|x|}}+3e^{\frac{-1}{x}}}{(a+2)e^{\frac{1}{|x|}}-e^{\frac{-1}{x}}}\right) & ;x\neq 0 \\ 0 & ;x=0 \end{cases}$ is differentiable at $x=0$, then $[a]=$_____ (where [.] is G.I.F.)

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